

What is this?

*This notebook is part of a collection of *pluto* notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.*

Table of Contents

Science Case: Sunspot Number

Nuclear fusion

The photosphere

Sunspots

Exercise: Sun's ~ 11 year cycle

Reference & Material

Further Material

Credits

Course Flow

TIME-DOMAIN ASTROPHYSICS

Stefano Covino
INAF / Brera Astronomical Observatory



Science Case: Sunspot Number

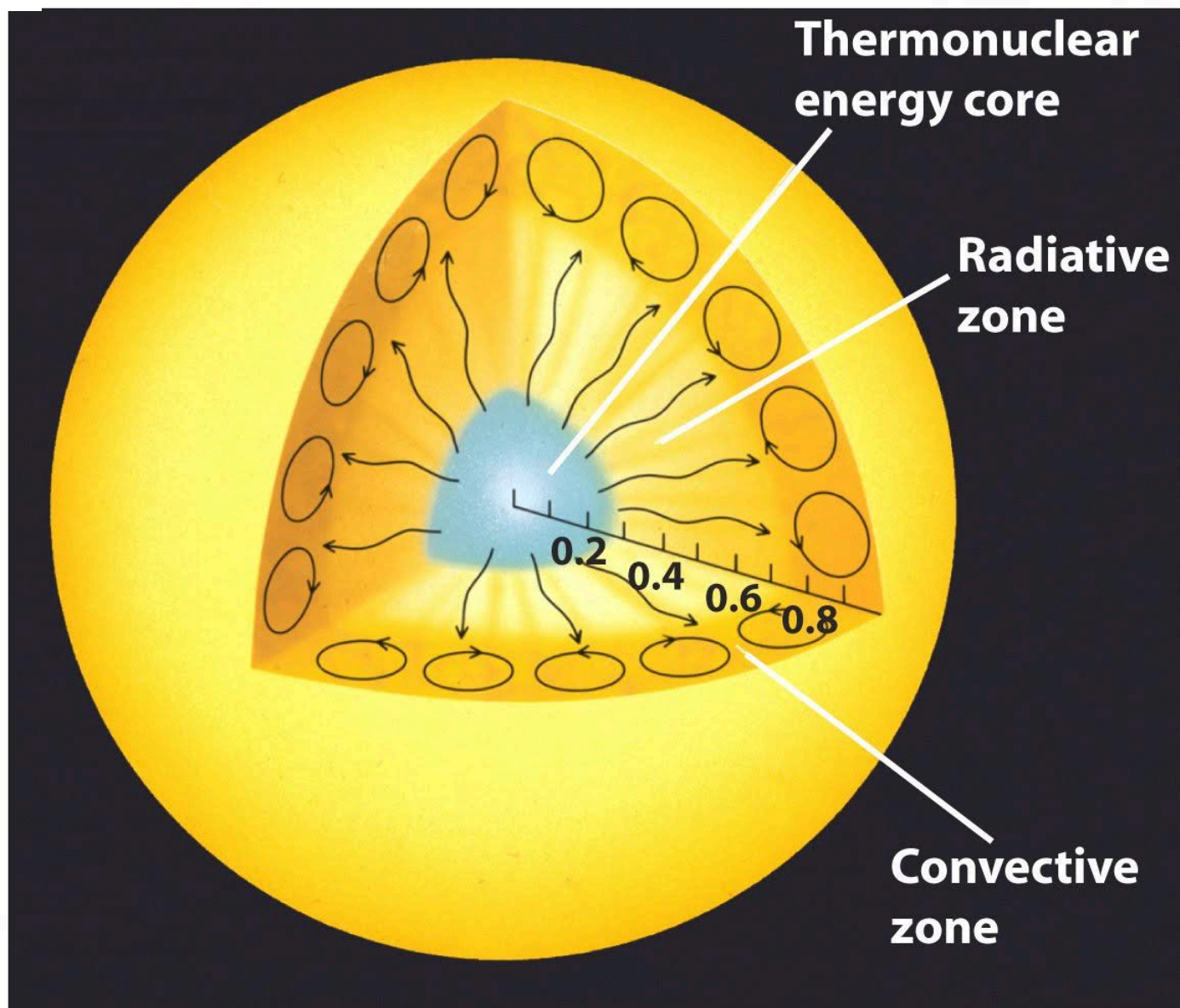
Nuclear fusion

.....

- The energy released by the Sun in a nuclear reaction corresponds to a slight reduction of mass according to Einstein's equation $E = mc^2$.
- Thermonuclear fusion occurs only at very high temperatures; for example, hydrogen fusion occurs only at temperatures in excess of about 10^7 K.
- In the Sun, fusion occurs only in the dense, hot core



- Hydrogen fusion takes place in a core extending from the Sun's center to about 0.25 solar radius.
- The core is surrounded by a radiative zone extending to about 0.71 solar radius.
- In this zone, energy travels outward through radiative diffusion.
- The radiative zone is surrounded by a rather opaque convective zone of gas at relatively low temperature and pressure. In this zone, energy travels outward primarily through convection.

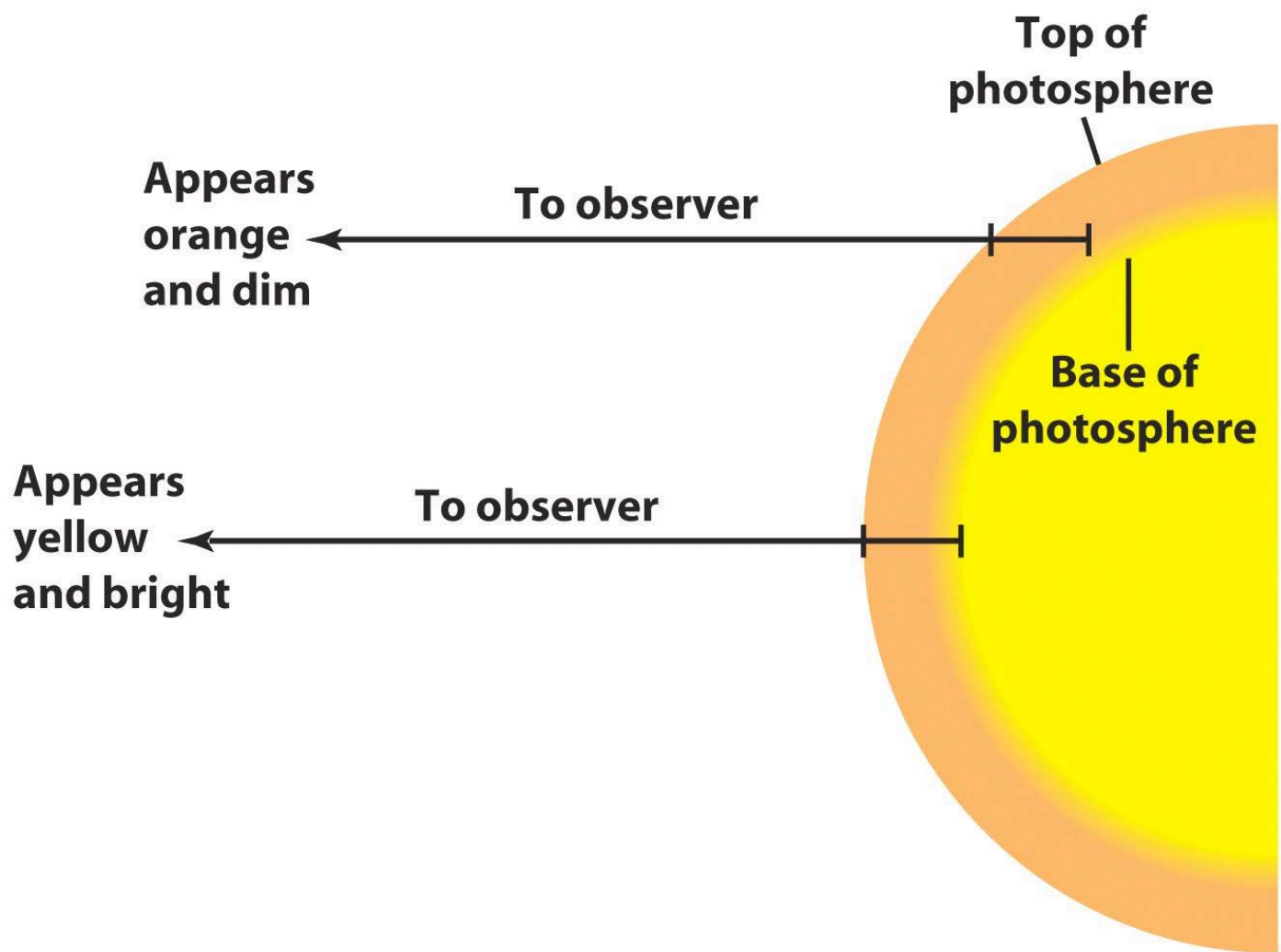


The photosphere

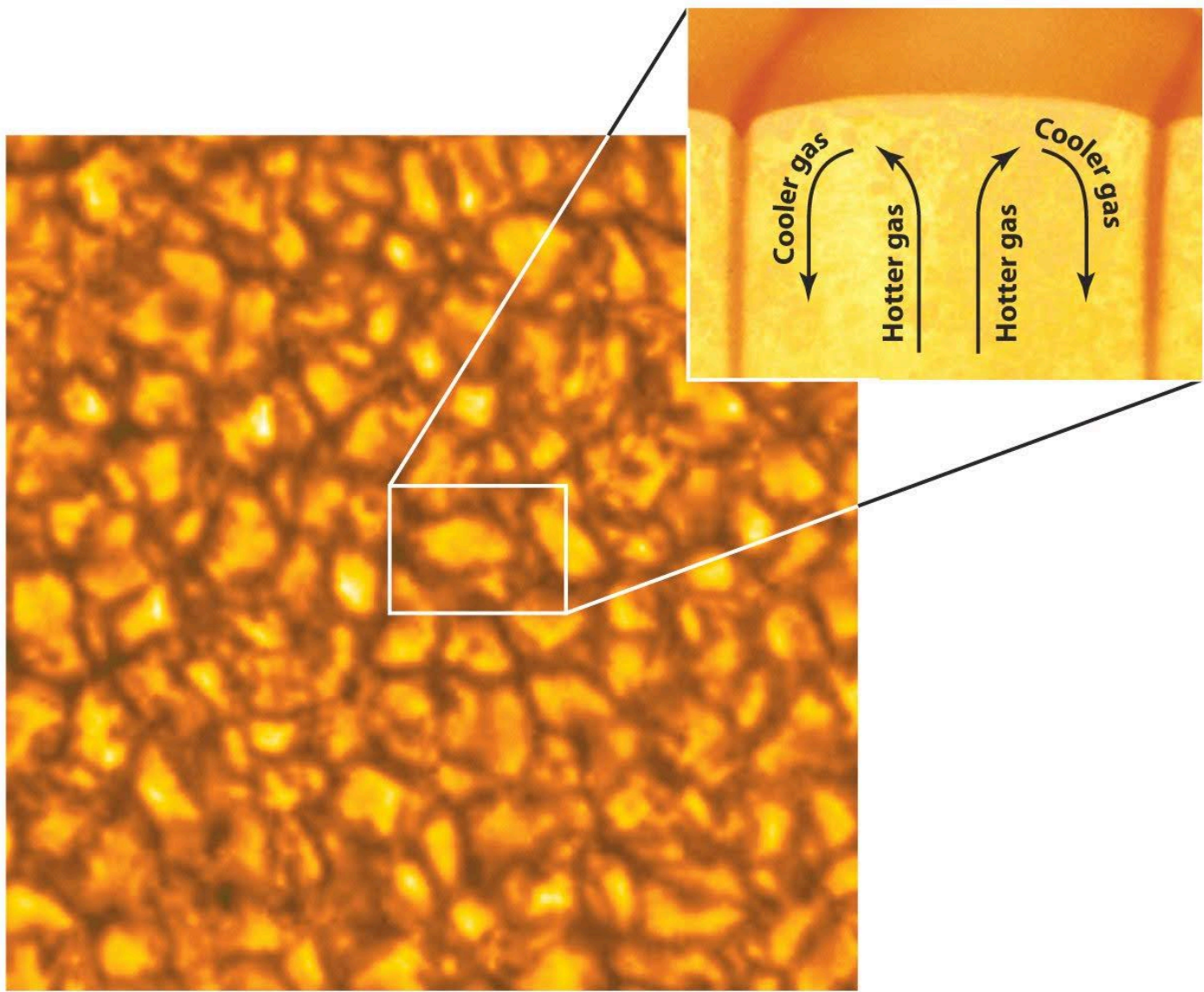
- The Sun's atmosphere has three main layers:
 - the photosphere
 - the chromosphere
 - the corona
- Everything below the solar atmosphere is called the solar interior.
- The visible surface of the Sun, the photosphere, is the lowest layer in the solar atmosphere.
- The spectrum of the photosphere is similar to that of a blackbody at a temperature of 5800 K



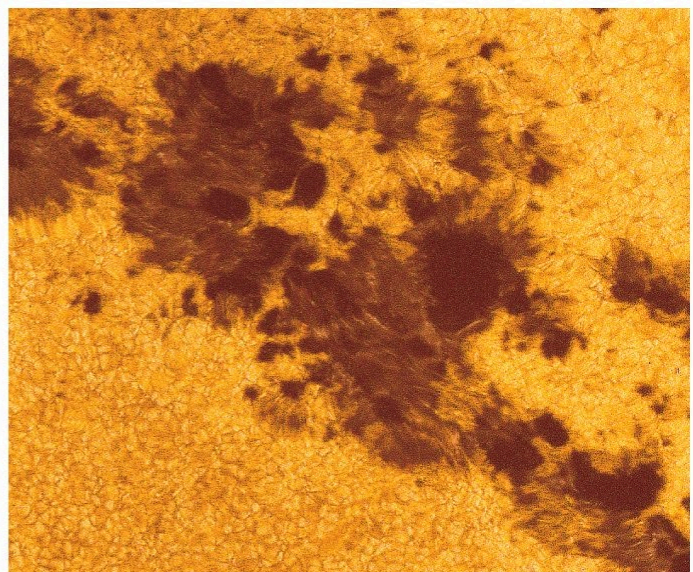
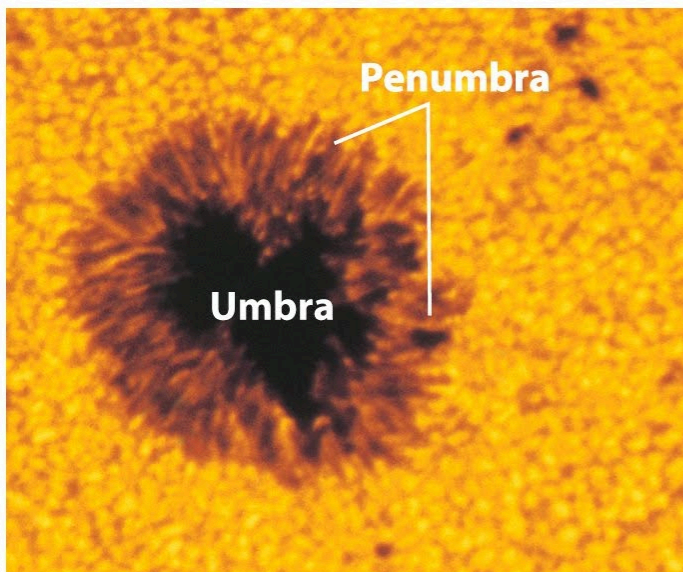
- Looking at the photosphere it is easy to observe a phenomenon known as *limb darkening*:



- The external layers of the the Sun are in convective equilibrium. And relatively high-resolution observations show the convective cells:



Sunspots



- Due to magnetic phenomena at the surface, occasionally it is possible to observe sunspots.
- The observations of persistent (for a few weeks) sunspot groups allow one to measure the Sun rotation period (and also the differential rotation).



November 9



November 12



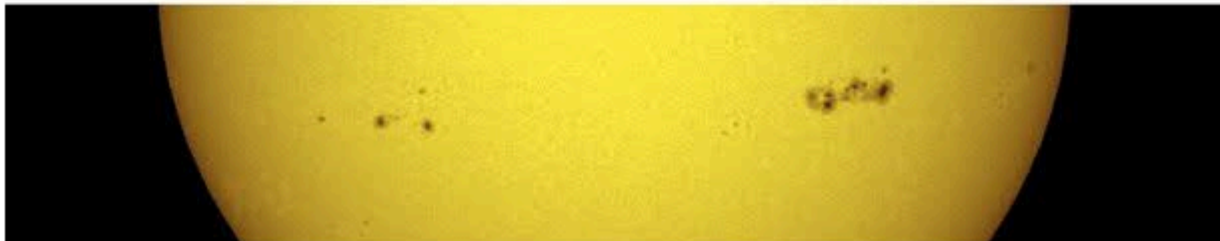
November 14



November 15

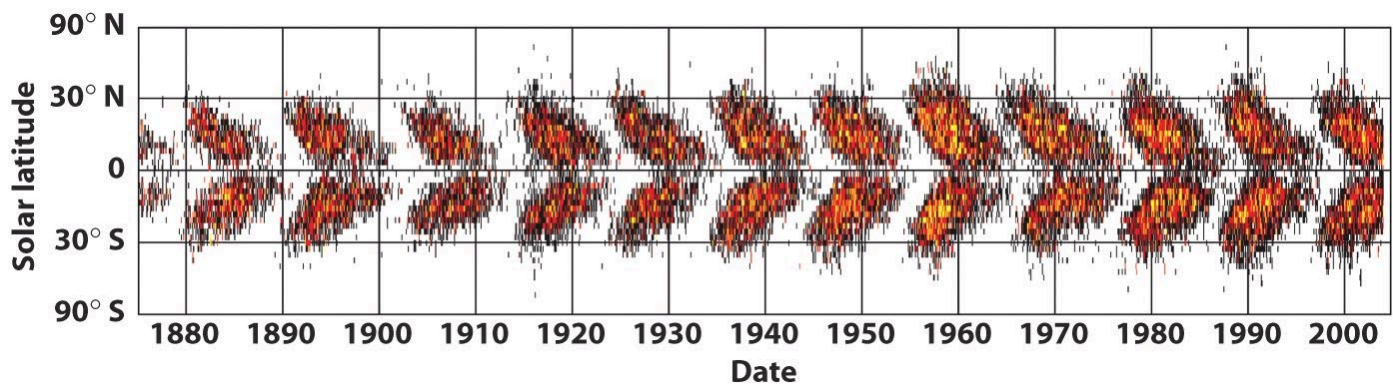
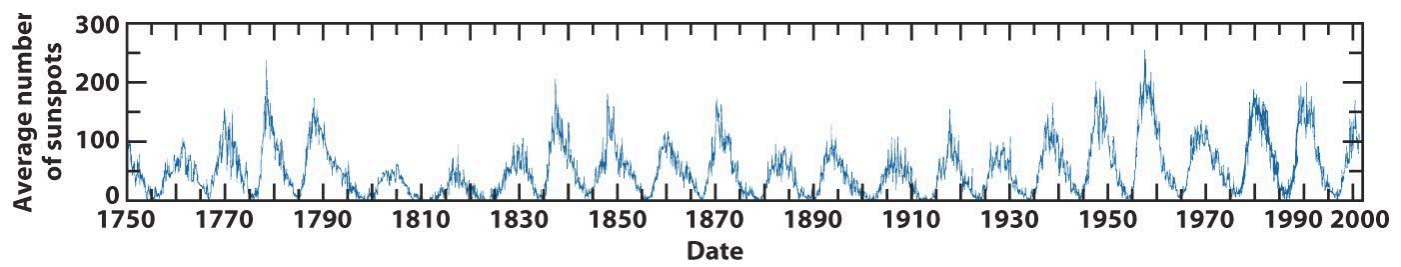


November 17



November 19

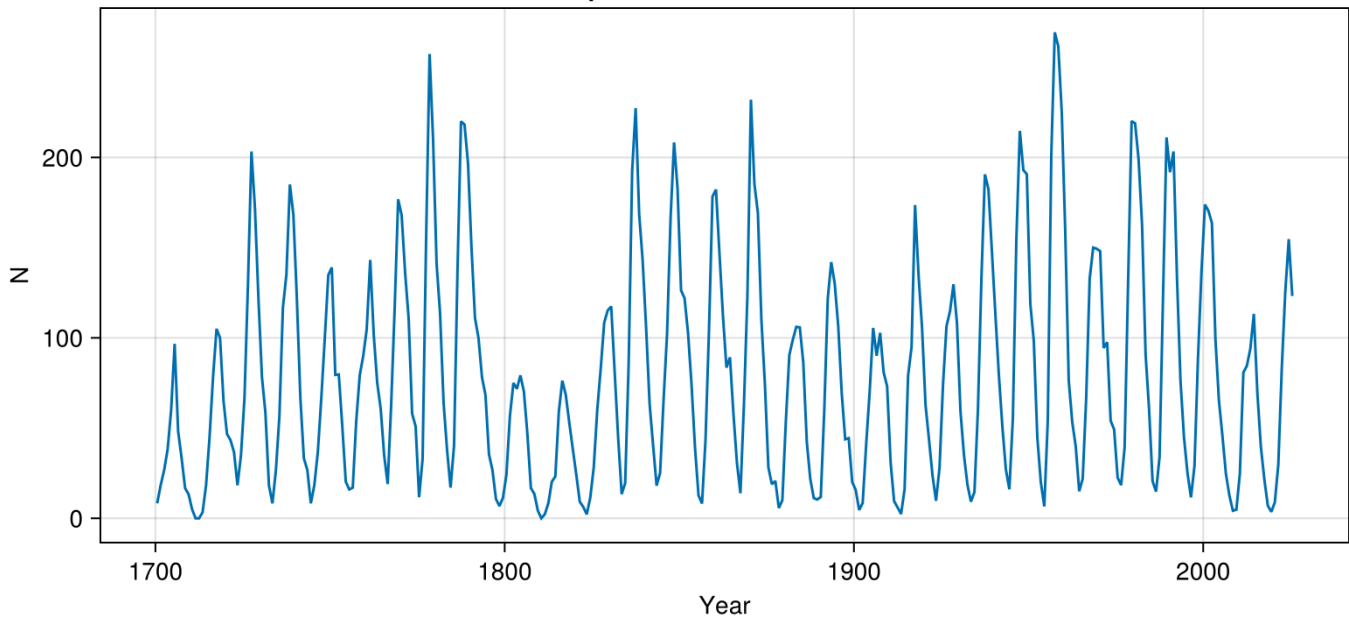
- It was soon realized that the visibility and number of sunspots follow a ~ 11 years cycle (in reality a ~ 22 magnetic cycle).



- The Sun's surface features vary in an 11-year cycle – the sunspot cycle.
- The average number of sunspots increases and decreases in a regular cycle of approximately 11 years, with reversed magnetic polarities from one 11-year cycle to the next.
- This is related to a 22-year cycle (the solar cycle) in which the surface magnetic field increases, decreases, and then increases again with the opposite polarity.
- Two sunspot cycles make up one 22-year solar cycle.

Exercise: Sun's ~ 11 year cycle

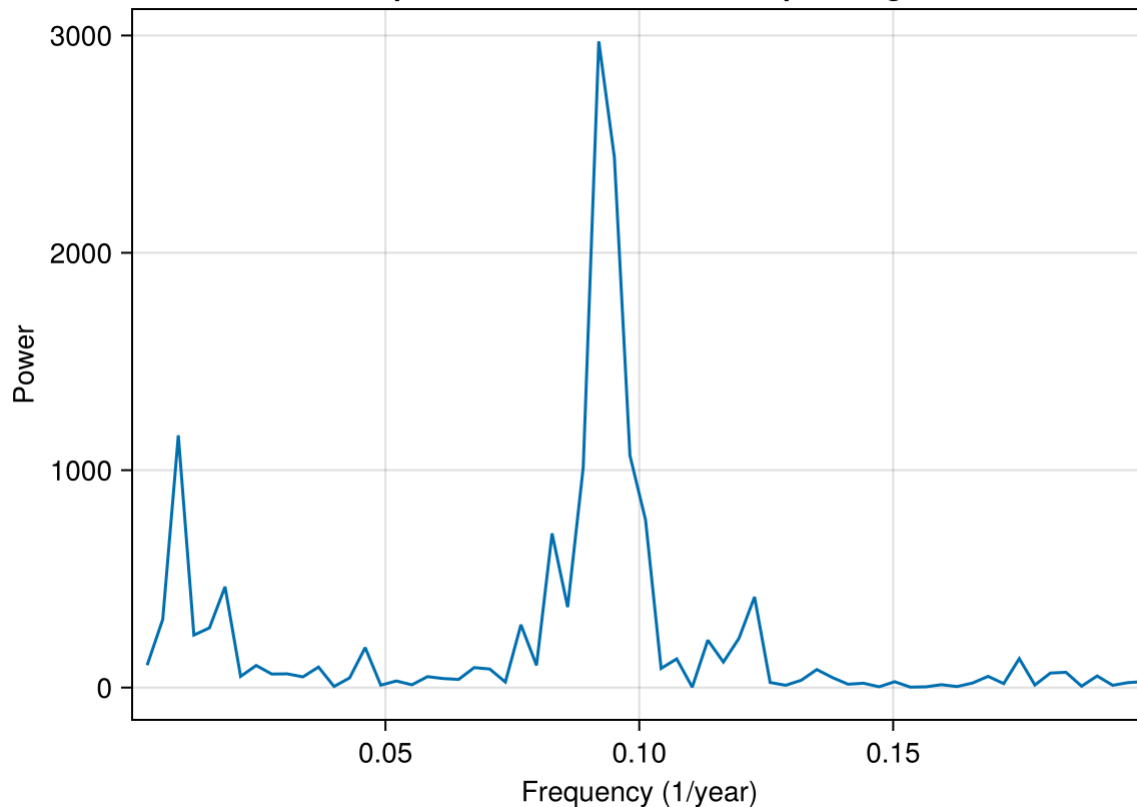
Sunspot number historical record



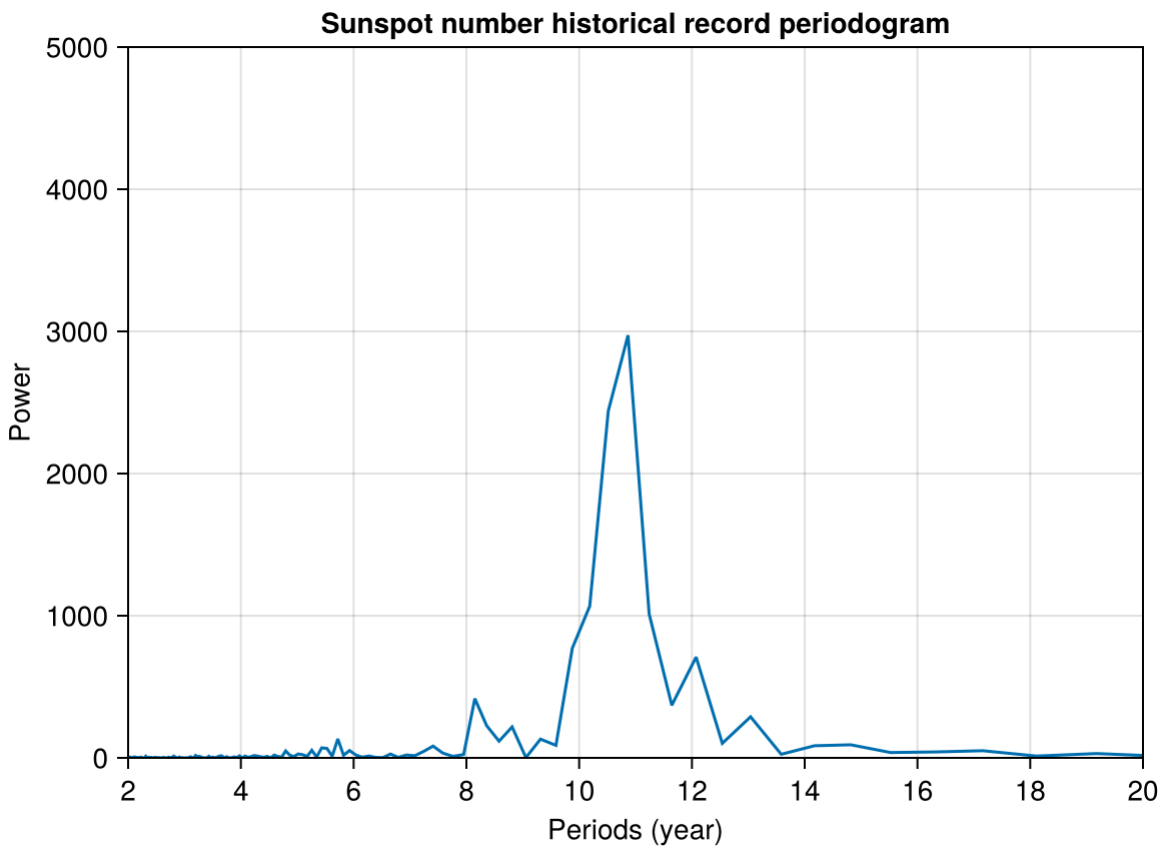
- A cyclicity is clearly visible, nevertheless cycles do not repeat identically.
- This implies that their power at different frequencies, although not necessarily with a strict periodicity.

Periods rather than frequencies? ☐

Sunspot number historical record periodogram



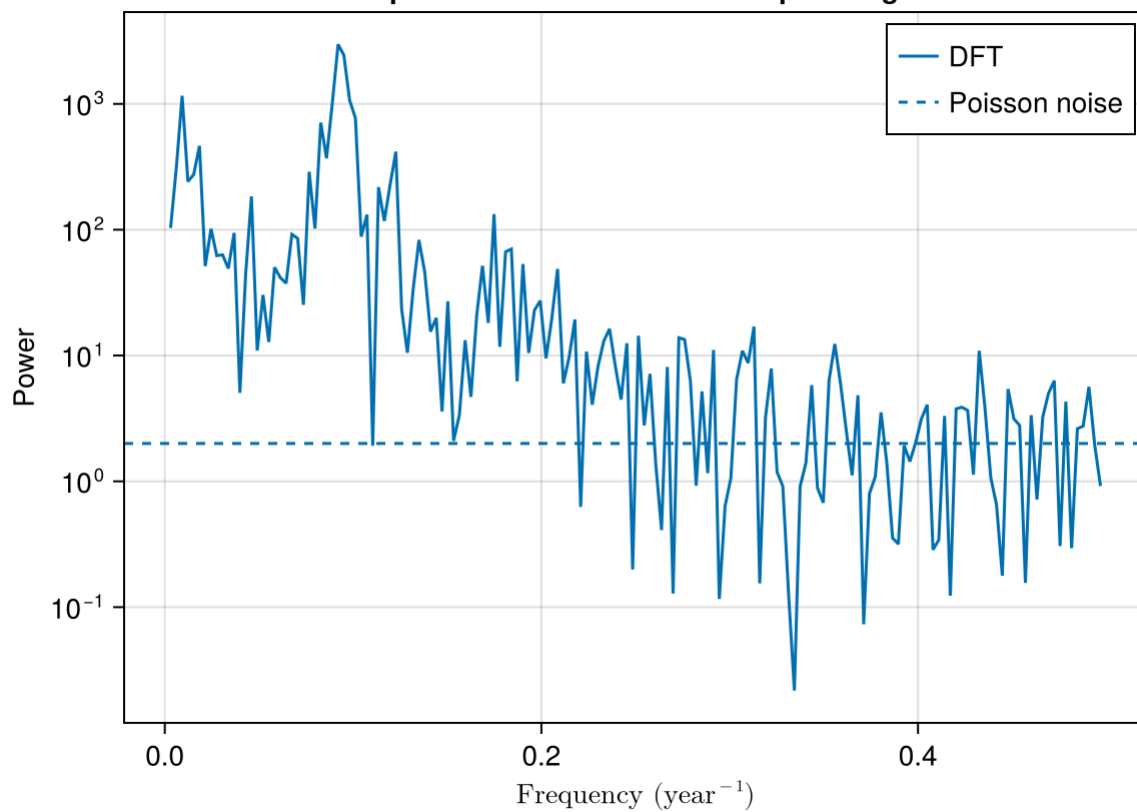
- A strong peak (or, as we are going to see, multiple peaks) is present at a ~ 11 year period, but there is power at a period of ~ 60 years and beyond.
- Let's see better the "peak" at ~ 11 years.



Period with maximum power: **11.2** years.

- As a matter of fact, the ~ 11 year periodicity is in reality a set of multiple peaks with periods ranging from approximately 8 to 12 years.

Sunspot number historical record periodogram



- A logarithmic plot shows a typical red-noise behavior that might be due to different reasons.
- In any case, this means that for computing the significance of a peak one unavoidably needs to model the underlying noise.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Norton et al. \(2023\) - "Solar Cycle Observations"](#)

Further Material

Papers or sites for examining more closely some of the discussed topics.

- ["Solar cycle progression"](#)

Credits

This notebook contains no external material.

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about spectral analysis	Science case about X-ray binaries	Course Summary
html	Lecture about spectral analysis	Science case about X-ray binaries	Course Summary

Copyright

This notebook is provided as [Open Educational Resource](#). Feel free to use the notebook for your own purposes. The text is licensed under [Creative Commons Attribution 4.0](#), the code of the examples, unless obtained from other properly quoted sources, under the [MIT license](#). Please attribute the work as follows: *Stefano Covino, Time Domain Astrophysics - Lecture notes featuring computational examples, 2026*.

Notebook v1.0.0 - 8 April 2026